Virtualization

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Virtualization: An Introduction (T123)

- Based on: "Cloud Computing: Theory and Practice" by Dan C. Marinescu
- Virtualization is a key technology in cloud computing.
- It separates hardware from software for better system efficiency.
- A 2009 Gartner report identified virtualization as a top strategic technology.

The Core Idea

- Traditional systems: Host OS directly manages hardware.
- Virtualization: Multiple guest OSs run on the same hardware, independent of the host OS.
- This is achieved by adding a *virtualization layer*.

Virtualization Layer: The Hypervisor

- The virtualization layer is also known as the *hypervisor* or *Virtual Machine Monitor (VMM)*.
- It manages the VMs.
- It converts real hardware into virtual hardware.
- Allows different OSs (e.g., Linux, Windows) to run simultaneously on the same physical machine.

Levels of Virtualization Implementation

- **Hardware Abstraction Level:** Virtualization directly on bare hardware.
  - Creates a virtual hardware environment for VMs.
  - Manages underlying hardware.
  - Aims to improve hardware utilization by multiple users.
  - Example: Xen hypervisor.

Levels of Virtualization Implementation (cont.)

- **Operating System Level:** Abstraction layer between OS and applications.
  - Creates isolated containers on a single physical server.
  - OS instances utilize the hardware.

Hypervisor Architecture

- Hypervisor sits directly between hardware and OS.
- Supports hardware-level virtualization.
- Manages CPU, memory, disk, and network interfaces.

Types of Virtualization Architectures

- **Hypervisor Architecture:** Described previously.
- **Paravirtualization:** Guest OS is modified to cooperate with the hypervisor.
- **Host-Based Virtualization:** Virtualization layer runs on top of a host OS.

Examples of Virtualization Technologies

- **VMware Workstation:** Host-based virtualization.
- **Xen:** Hypervisor for various architectures (IA-32, x86-64, Itanium, PowerPC). Can modify Linux to act as a hypervisor.
- **KVM (Kernel-based Virtual Machine):** Linux kernel virtualization infrastructure. Supports hardware-assisted virtualization and paravirtualization.

KVM and VirtIO

- KVM uses Intel VT-x or AMD-v for hardware-assisted virtualization.
- It uses VirtIO framework for paravirtualization.
- VirtIO includes:
  - Paravirtual Ethernet card
  - Disk I/O controller
  - Balloon device (memory management)
  - VGA graphics interface

Benefits of Virtualization

- Improved resource utilization.
- Application flexibility.
- Better system efficiency.
- Enlarged memory space (virtual memory).
- Enhanced use of compute engines, networks, and storage.